

UNVEILING ACOUSTIC EVENT REALIZATION HEADS IN TEXT-TO-AUDIO GENERATIVE MODELS

Hyewon Cho¹, Sunghyun Ryu², Won-Gook Choi¹, and Joon-Hyuk Chang^{1,2,†}

¹Department of Electronic Engineering, Hanyang University, Seoul, Republic of Korea

²Department of Artificial Intelligence, Hanyang University, Seoul, Republic of Korea
{haewonn0204, rsh001002, onlyworld94, jchang}@hanyang.ac.kr

ABSTRACT

Recent text-to-audio (TTA) models have advanced toward generating diverse and realistic audio from text descriptions. In these models, cross-attention is commonly used to incorporate textual information during audio generation, yet our understanding of individual cross-attention heads remains limited. We therefore refer to the generation of acoustic events described in the input caption as acoustic event realization (AER) and term cross-attention heads important for this process AER heads. To identify these heads, we introduce event attention mass (EAM), which measures the attention each head allocates to acoustic-event tokens. Across three TTA models, we rank cross-attention heads using EAM, identify the top-ranked heads as AER heads, and evaluate their influence through head masking. AER heads showed a stronger influence than random or bottom-ranked heads on both acoustic event generation and text-to-audio generation performance, while low-ranked head masking largely preserved generation performance. These findings provide insight into the functional roles of individual cross-attention heads in TTA models and may support future research on acoustic-event control and the identification of other functional heads. Audio samples are available at <https://hyewon24.github.io/aer-head-demo/>.

Index Terms— text-to-audio generation, acoustic event realization heads, cross-attention, event attention mass

1. INTRODUCTION

Recent text-to-audio (TTA) models have advanced toward generating diverse and realistic audio from natural-language descriptions [1–3]. Among them, diffusion-based models have become widely adopted in modern TTA systems [4–7]. These models often use cross-attention to incorporate fine-grained textual information into audio representations during generation [3, 5, 6]. Given its important role in text conditioning, recent studies have explored cross-attention for analysis and generation control. Auffusion [8] visualizes cross-attention maps for text-audio alignment, PPAAE [9] exploits them for precise audio editing, and TAG [10] manipulates cross-attention for temporal control of acoustic events. Although cross-attention consists of multiple heads that can exhibit different attention patterns and distinct functional roles, our understanding of these heads in TTA models remains limited.

Meanwhile, studies in other domains have identified attention heads with specialized functions. In image generation, concept-specific cross-attention heads can be manipulated to control corresponding visual concepts [11]. Similarly, retrieval heads in large

language models (LLMs) and optical character recognition (OCR) heads in large vision-language models have been identified [12, 13].

Motivated by these studies, we investigate cross-attention heads important for acoustic event realization (AER), which we define as the generation of acoustic events described in the input caption. We term these heads AER heads. To identify AER heads, we propose event attention mass (EAM) as a measure of the attention each head allocates to acoustic-event tokens, following prior work showing that token-level attention is important for audio generation and control [10, 14], and that functional heads can be identified by examining attention related to specific tasks [11–13]. We then use EAM to rank cross-attention heads across three TTA models and identify the top-ranked heads as AER heads.

To evaluate the influence of AER heads, we compared the effects of masking AER heads, randomly selected heads, and bottom-ranked heads. Compared with random and bottom-ranked head masking, masking AER heads caused greater degradation in acoustic event generation. Furthermore, AER heads had a stronger influence on text-to-audio generation performance than random and bottom-ranked heads, while low-ranked head masking largely preserved performance.

Our main contributions are as follows:

- We introduce EAM for identifying AER heads based on their attention to acoustic-event tokens.
- AER heads are shown to have a stronger influence than other heads on the generation of acoustic events described in the input caption.

2. BACKGROUND

2.1. Cross-Attention in Diffusion-based TTA Models

Many diffusion-based TTA models use multi-head cross-attention to incorporate fine-grained textual information during generation. Earlier models such as AudioLDM [4] employed CLAP [15] embeddings as conditioning for cross-attention, providing a global semantic representation of the prompt. In contrast, recent models use token-level language-model embeddings such as FLAN-T5 [16], which preserve individual token information and interact with audio representations through cross-attention.

This form of conditioning has been adopted in various diffusion-based TTA architectures. AudioLDM2 [5] and TANGO [3] use UNet-based architectures with FLAN-T5 token-level conditioning. In contrast, stable audio open (SAO) [6] uses a diffusion transformer [17] backbone with cross-attention conditioned on T5 [18] text representations. Despite these differences, all three models

[†] Joon-Hyuk Chang is the corresponding author.

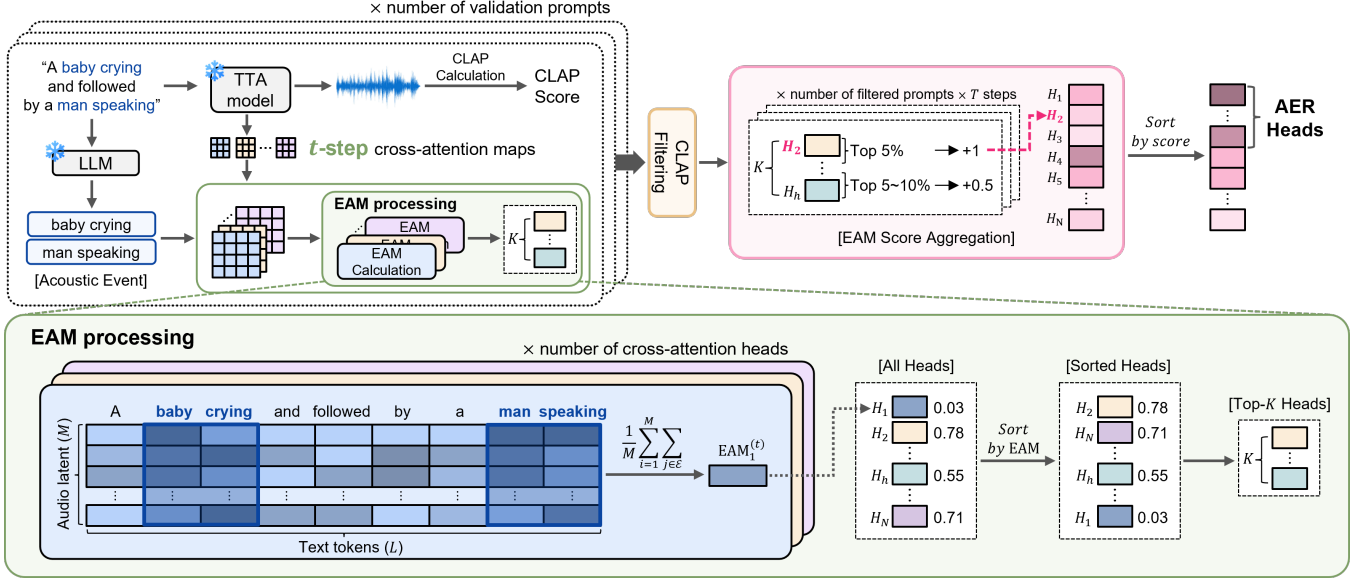


Fig. 1. Overview of the proposed AER head identification framework using an example prompt. The enlarged EAM processing block shows the computation at denoising step t , with H_h denoting the h -th cross-attention head.

use cross-attention to incorporate textual information into audio generation.

2.2. Cross-Attention Manipulation and Functional Heads

Cross-attention has been utilized for generation control. A representative example is TAG [10], which achieves temporal control of acoustic events through cross-attention manipulation. Specifically, it applies positive biases to cross-attention logits within the target time interval and negative biases outside it, concentrating event-related attention on the desired temporal region.

Beyond such cross-attention manipulation, studies in other domains have shown that individual attention heads can exhibit different functional roles [13, 19–22]. For example, retrieval heads [12] in LLMs play an important role in retrieving relevant information from long contexts. Retrieval heads are identified by scoring how often each head attends to retrieval-relevant input information when the model generates the corresponding correct output. Masking these heads significantly degrades retrieval performance, validating their functional role.

3. METHOD

In this section, we present the framework for AER head identification, and the overall identification process is illustrated in Fig. 1. AER heads are identified by ranking heads based on EAM. We first introduce EAM, then describe CLAP-based filtering used to exclude generations with low text-audio relevance, and finally present the EAM-based head ranking procedure for AER head identification.

3.1. Event Attention Mass (EAM)

For AER head identification, we define EAM to quantify the attention each cross-attention head allocates to acoustic-event tokens. We treat audio as a composition of acoustic events and use an LLM to extract sound-related descriptions from the caption as acoustic-event

phrases. For example, “A baby crying and followed by a man speaking” is decomposed into “baby crying” and “man speaking.” For an original caption of L tokens and an audio latent sequence of length M , let $\mathcal{E} \subseteq \{1, \dots, L\}$ and $A^{(h,t)} \in \mathbb{R}^{M \times L}$ be the set of token indices corresponding to the extracted acoustic-event phrases and the cross-attention map of head h at denoising step t , respectively. Then, EAM is defined as follows:

$$\text{EAM}_h^{(t)} = \frac{1}{M} \sum_{i=1}^M \sum_{j \in \mathcal{E}} A_{i,j}^{(h,t)}, \quad (1)$$

where $A_{i,j}^{(h,t)}$ denotes the attention score between the i -th audio latent and the j -th text token. Accordingly, a higher EAM indicates that the head allocates more attention mass to acoustic-event tokens.

3.2. CLAP-based Filtering

To reduce the influence of generations with low text-audio relevance on head scoring, we apply CLAP-based filtering before EAM score aggregation. For each model, we compute the CLAP similarity between the generated audio and the input caption and exclude the bottom 10% of generations by CLAP score. Only the remaining generations are used for AER head scoring.

3.3. AER Head Identification

We hypothesize that attention heads with high EAM are more strongly involved in generating acoustic events, since they attend more strongly to tokens corresponding to acoustic events. Based on this hypothesis, we identify heads that consistently exhibit relatively high EAM across different denoising steps and generations as AER heads.

For each audio sample, EAM values are computed for all cross-attention heads at each denoising step. We then rank the heads by EAM and assign tiered scores to the top- K heads, where K denotes the number of heads in the top 10%. More specifically, the score w_h^t is defined as:

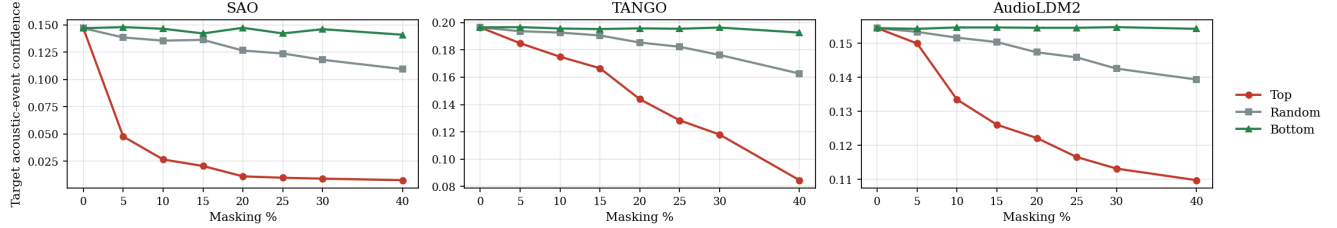


Fig. 2. Change in TAEC as the masking ratio increases for top-ranked, random, and bottom-ranked heads. A larger reduction in TAEC indicates that the masked heads have a stronger influence on target acoustic-event realization.

$$w_h^t = \begin{cases} 1.0, & r_h^t \leq 0.05N, \\ 0.5, & 0.05N < r_h^t \leq 0.10N, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

where N and r_h^t denote the total number of heads and the EAM rank of head h at step t , respectively. This tiered scoring gives greater weight to the highest-ranked heads while still considering other highly ranked heads. The assigned scores are then accumulated for each head across denoising steps and generations retained after CLAP-based filtering to obtain its final EAM score. We refer to accumulating these tiered scores as rank aggregation, and the resulting EAM scores indicate how consistently each head ranks highly across denoising steps and generations. We consider top-ranked heads based on their final EAM scores as AER heads and evaluate their masking effects at multiple masking ratios rather than imposing a universal cutoff.

4. EXPERIMENTS

Datasets. For AER head identification, we used 494 audio clips from the AudioCaps [23] validation set, each with five captions. We used the first two captions per clip (988 total) and extracted acoustic-event phrases using Qwen3-4B-Instruct-2507 [24]. To reduce generation failures from multi-event prompts, we retained prompts containing at most two acoustic events, yielding 594 prompts. For each model, we then excluded the bottom 10% of generations by CLAP similarity [15], leaving 534 generations for AER head scoring.

For evaluation, we used 957 samples from the AudioCaps test set, using the first of five captions as the generation prompt. To quantify target acoustic-event realization based on AudioSet [25] class confidence, each target event was mapped to the most semantically relevant AudioSet class. Qwen3-4B-Instruct-2507 was used to extract acoustic-event phrases from each prompt and perform the mapping, and three samples without an appropriate class were excluded, resulting in 954 samples.

Models and Generation Settings. We conducted experiments on three diffusion-based TTA models: SAO [6], TANGO [3], and AudioLDM2 [5], with sampling rates of 44.1 kHz for SAO and 16 kHz for the other two models. SAO and TANGO contain 576 and 195 cross-attention heads, respectively, while for AudioLDM2, we analyzed only the 128 FLAN-T5 [16]-conditioned heads that preserve token-level correspondence with the input caption. All models generated 10-s audio using 100 inference steps, with the guidance scale set to 7.0 for SAO and 3.0 for TANGO and AudioLDM2.

Evaluation Protocol and Metrics. To evaluate the influence of AER heads, we compared AER head masking with random and bottom-ranked head masking. Specifically, we masked the selected heads by zeroing their cross-attention outputs, and for random masking, heads were independently sampled for each sample. We assessed these effects on acoustic event generation and text-to-audio

generation performance. We define target acoustic-event confidence (TAEC) as the classifier score of the corresponding AudioSet class predicted by a pretrained PANNs [26], with higher confidence indicating stronger target-event presence. For prompts containing multiple target events, TAEC is computed as the sum of the corresponding class confidences. To evaluate text-to-audio generation performance, we used Fréchet Audio Distance (FAD) based on VGGish [27] and CLAP similarity (LAION-CLAP [15]), which measure audio quality and text-audio relevance, respectively.

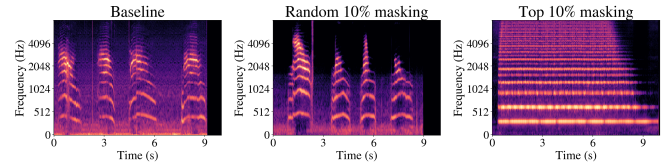


Fig. 3. Mel-spectrograms of audio generated by SAO for “A cat meowing repeatedly” under baseline, random 10%, and top 10% masking. Repeated cat meowing is retained in the baseline and random masking, while top-ranked masking produces an instrument-like sound instead.

5. RESULTS AND ANALYSIS

5.1. Influence of AER Heads on Text-to-Audio Generation

To examine the influence of AER heads on acoustic event generation, we compared top-ranked, random, and bottom-ranked head masking, with the resulting TAEC for each model shown in Fig. 2. Across all three models, masking top-ranked heads resulted in a larger reduction in TAEC than masking the same proportion of random or bottom-ranked heads. Fig. 3 shows an example of this result, where the target acoustic event was preserved under the baseline and random masking conditions but disappeared and was replaced by other sounds under top-ranked head masking. These results show that AER heads have a stronger influence than other heads on generating acoustic events described in the input caption.

We further evaluated the influence of AER heads on text-to-audio generation performance using FAD and CLAP. As shown in Table 1, masking top-ranked heads resulted in greater degradation in FAD and CLAP than masking random or bottom-ranked heads across all three models. These findings show that AER heads have a stronger influence than other heads on text-to-audio generation performance.

We also examined the effect of progressively masking low-ranked heads. Across all three models, FAD and CLAP remained close to the baseline up to 70% for SAO, 20% for TANGO, and 40% for AudioLDM2, as shown in Table 2, while performance gradually degraded beyond these ranges. These results suggest that low-ranked heads have relatively limited influence on text-to-audio generation performance.

Table 1. FAD and CLAP under top-ranked, random, and bottom-ranked cross-attention head masking.

Metric		SAO			TANGO			AudioLDM2		
FAD↓	Baseline	3.535			3.045			4.116		
	Masking	Top	Random	Bottom	Top	Random	Bottom	Top	Random	Bottom
	5%	7.427	3.550	3.345	3.136	3.117	3.010	4.175	4.148	4.118
	10%	7.864	3.535	3.314	3.277	3.095	3.011	4.426	4.161	4.121
	20%	13.693	3.830	3.255	5.398	3.519	2.993	4.924	4.263	4.091
	40%	21.623	3.972	3.411	10.923	4.934	3.197	5.521	4.471	4.083
CLAP↑	Baseline	0.282			0.499			0.422		
	Masking	Top	Random	Bottom	Top	Random	Bottom	Top	Random	Bottom
	5%	0.109	0.277	0.281	0.489	0.494	0.499	0.411	0.417	0.422
	10%	0.050	0.272	0.276	0.474	0.494	0.498	0.387	0.415	0.423
	20%	-0.001	0.252	0.282	0.431	0.481	0.496	0.369	0.408	0.423
	40%	-0.017	0.221	0.273	0.309	0.446	0.487	0.345	0.391	0.421

Table 2. FAD and CLAP under progressive low-ranked head masking. Results are shown around the masking range where these metrics remain largely preserved for each model.

Mask.	Model								
	SAO		TANGO			AudioLDM2			
	FAD↓	CLAP↑	Mask.	FAD↓	CLAP↑	Mask.	FAD↓	CLAP↑	
Base	3.535	0.282	Base	3.045	0.499	Base	4.116	0.422	
60%	<u>3.417</u>	0.272	10%	<u>3.011</u>	<u>0.498</u>	30%	<u>4.091</u>	0.423	
70%	3.196	<u>0.279</u>	20%	2.993	0.496	40%	4.083	0.421	
80%	3.382	0.268	30%	3.105	0.493	50%	4.150	0.418	

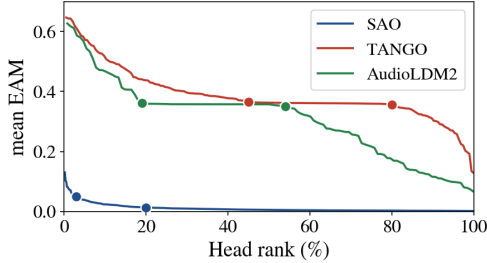
**Fig. 4.** Mean EAM of cross-attention heads across denoising steps and generations, ordered by the proposed ranking, for SAO, TANGO, and AudioLDM2. Markers indicate the qualitative boundaries between high-, middle-, and low-EAM regions used for interpretation.

Fig. 4 shows the mean EAM across the head ranking for each model. The model-specific masking effect patterns described above were associated with differences in their respective EAM distributions. For interpretation, we qualitatively divided each model’s EAM curve into high-, middle-, and low-EAM regions based on the relative EAM within that model. In SAO, the high-EAM region was concentrated within a small set of top-ranked heads, consistent with the large degradation observed at low masking ratios. In contrast, TANGO and AudioLDM2 exhibited broader high-EAM regions. In these models, the degradation increased gradually as masking expanded across the ranking, suggesting that a larger set of heads was involved in AER. In addition, the low-ranked masking ranges where performance remained close to the baseline were associated with the low-EAM regions, suggesting that low-EAM heads have relatively limited influence on text-to-audio generation performance. Overall, this relationship suggests that the EAM distribution reflects how broadly heads more strongly involved in AER are distributed across the ranking, consistent with the model-specific masking effects.

Table 3. Ablation results for the components of the AER head identification method. Values denote TAEC after masking the top 10% of heads identified by each variant; lower values indicate that the identified heads are more strongly involved in acoustic event realization.

Method	CLAP	Rank	Tiered	SAO	TANGO	AudioLDM2
Full	✓	✓	✓	0.0265	0.1750	0.1335
w/o CLAP Filtering		✓	✓	0.0288	0.1750	0.1335
w/o Tiered Scoring	✓		✓	0.0302	0.1750	0.1365
w/o Rank Aggregation	✓			0.0304	0.1768	0.1392

5.2. Ablation Study

To further examine the contribution of the components of the proposed method, we compared three variants: (1) without CLAP filtering, using all generations for scoring; (2) without tiered scoring, assigning a score of 1 to all heads in the top 10%; and (3) without rank aggregation, directly averaging raw EAM values across denoising steps and generations. We then masked the top 10% of heads identified by each variant and measured TAEC. As shown in Table 3, the proposed method achieved the minimum TAEC across all three models. This indicates that the proposed method more effectively identifies heads that have a stronger influence on the generation of acoustic events described in the input caption. Among the components, rank aggregation provided the most consistent contribution, as removing it increased TAEC across all three models. This supports using relative EAM ranks instead of raw EAM values, whose scales can vary across prompts and generations. Tiered scoring and CLAP filtering also contributed to the identification, although their effects varied across models.

6. CONCLUSION

This work identifies AER heads, cross-attention heads important for the generation of acoustic events described in input captions, using EAM across three TTA models. Across all three models, masking AER heads caused greater degradation in both acoustic event generation and text-to-audio generation performance than random or bottom-ranked head masking, while low-ranked head masking largely preserved performance. Moreover, the model-specific masking effects were consistent with differences in the EAM distributions, reflecting how broadly heads more strongly involved in AER are distributed across the ranking. These findings may support future research on acoustic-event control using AER heads and the identification of other functional heads in TTA models.

7. COMPLIANCE WITH ETHICAL STANDARDS

This study used publicly available datasets and pretrained models and did not involve human subjects; therefore, no ethical approval was required.

8. REFERENCES

- [1] F. Kreuk *et al.*, “AudioGen: Textually guided audio generation,” in *Proc. International Conference on Learning Representations (ICLR)*, 2023.
- [2] R. Huang *et al.*, “Make-An-Audio: Text-to-audio generation with prompt-enhanced diffusion models,” in *Proc. International Conference on Machine Learning (ICML)*, 2023, pp. 13916–13932.
- [3] D. Ghosal, N. Majumder, A. Mehrish, and S. Poria, “Text-to-audio generation using instruction guided latent diffusion model,” in *Proc. ACM International Conference on Multimedia (ACM MM)*, 2023, pp. 3590–3598.
- [4] H. Liu *et al.*, “AudioLDM: Text-to-audio generation with latent diffusion models,” in *Proc. International Conference on Machine Learning (ICML)*, 2023, pp. 21450–21474.
- [5] H. Liu *et al.*, “AudioLDM 2: Learning holistic audio generation with self-supervised pretraining,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 2871–2883, 2024.
- [6] Z. Evans *et al.*, “Stable Audio Open,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5.
- [7] S.-G. Lee *et al.*, “ETTA: Elucidating the design space of text-to-audio models,” in *Proc. International Conference on Machine Learning (ICML)*, 2025, pp. 33157–33180.
- [8] J. Xue, Y. Deng, Y. Gao, and Y. Li, “Auffusion: Leveraging the power of diffusion and large language models for text-to-audio generation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 4700–4712, 2024.
- [9] M. Xu *et al.*, “Prompt-guided precise audio editing with diffusion models,” in *Proc. International Conference on Machine Learning (ICML)*, 2024, pp. 55126–55143.
- [10] H. Zhang *et al.*, “TAG: Structured temporal audio generation via LLM-guided manual scription and control,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026, pp. 17752–17756.
- [11] J. Park *et al.*, “Cross-attention head position patterns can align with human visual concepts in text-to-image generative models,” in *Proc. International Conference on Learning Representations (ICLR)*, 2025.
- [12] W. Wu *et al.*, “Retrieval head mechanistically explains long-context factuality,” in *Proc. International Conference on Learning Representations (ICLR)*, 2025, pp. 33762–33775.
- [13] I. Baek, H. Chang, S. Ryu, and H. Lee, “How do large vision-language models see text in image? unveiling the distinctive role of OCR heads,” in *Proc. Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025, pp. 20441–20453.
- [14] H. Kang, G. Han, Y. Jeong, and H. Park, “AudioGenX: Explainability on text-to-audio generative models,” in *Proc. AAAI Conference on Artificial Intelligence (AAAI)*, 2025, pp. 17733–17741.
- [15] Y. Wu *et al.*, “Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5.
- [16] H. W. Chung *et al.*, “Scaling instruction-finetuned language models,” *Journal of Machine Learning Research*, vol. 25, no. 70, pp. 1–53, 2024.
- [17] W. Peebles and S. Xie, “Scalable diffusion models with transformers,” in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 4172–4182.
- [18] C. Raffel *et al.*, “Exploring the limits of transfer learning with a unified text-to-text transformer,” *Journal of Machine Learning Research*, vol. 21, no. 140, pp. 1–67, 2020.
- [19] E. Todd *et al.*, “Function vectors in large language models,” in *Proc. International Conference on Learning Representations (ICLR)*, 2024.
- [20] A. K. Singh *et al.*, “What needs to go right for an induction head? a mechanistic study of in-context learning circuits and their formation,” in *Proc. International Conference on Machine Learning (ICML)*, 2024, pp. 45637–45662.
- [21] H. Yuan *et al.*, “Whispers that shake foundations: Analyzing and mitigating false premise hallucinations in large language models,” in *Proc. Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024, pp. 2670–2683.
- [22] K. Yin and J. Steinhardt, “Which attention heads matter for in-context learning?,” in *Proc. International Conference on Machine Learning (ICML)*, 2025, pp. 72428–72461.
- [23] C. D. Kim, B. Kim, H. Lee, and G. Kim, “AudioCaps: Generating captions for audios in the wild,” in *Proc. the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2019, pp. 119–132.
- [24] A. Yang *et al.*, “Qwen3 technical report,” *arXiv preprint arXiv:2505.09388*, 2025.
- [25] J. F. Gemmeke *et al.*, “Audio Set: An ontology and human-labeled dataset for audio events,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017, pp. 776–780.
- [26] Q. Kong *et al.*, “PANNs: Large-scale pretrained audio neural networks for audio pattern recognition,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 2880–2894, 2020.
- [27] K. Kilgour, M. Zuluaga, D. Roblek, and M. Sharifi, “Fréchet audio distance: A metric for evaluating music enhancement algorithms,” *arXiv preprint arXiv:1812.08466*, 2018.